Day 043 | Unit 03 | Teacher plan

Integers and Rational Numbers

Find Horizontal and Vertical Distance

Standards: 6.NS.C.8

Learning target: I can graph points and find horizontal or vertical distance.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Plot both points. If x-coordinates match, count the vertical units; if y-coordinates match, count horizontal units. On opposite sides of zero, add distances to zero. On the same side, subtract magnitudes.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 7

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01 / Learn

Plot both points. If x-coordinates match, count the vertical units; if y-coordinates match, count horizontal units. On opposite sides of zero, add distances to zero. On the same side, subtract magnitudes.

Worked example 1

Plot (-4, 4) and (-8, 4). Find their distance.

1. Plot both points.

2. If x-coordinates match, count the vertical units; if y-coordinates match, count horizontal units.

3. Only one coordinate changes.

4. Same side of 0: subtract magnitudes to get 4.

5. State the result in the form requested: 4 units.

Answer: 4 units

Worked example 2

Plot A(-7, -4), B(2, -4), and C(-7, 4). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.

1. Plot both points.

2. AB is horizontal: 7 + 2 = 9.

3. AC is vertical: 4 + 4 = 8.

4. Each segment crosses an axis.

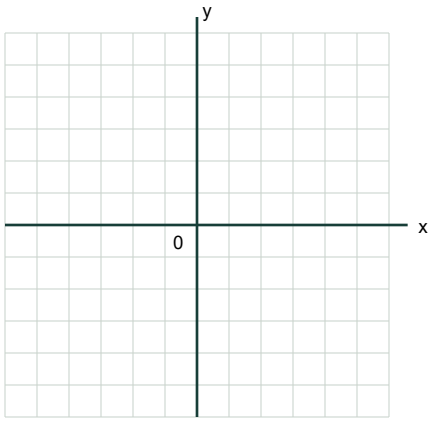
Answer: AB = 9 units; AC = 8 units

Day 043 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

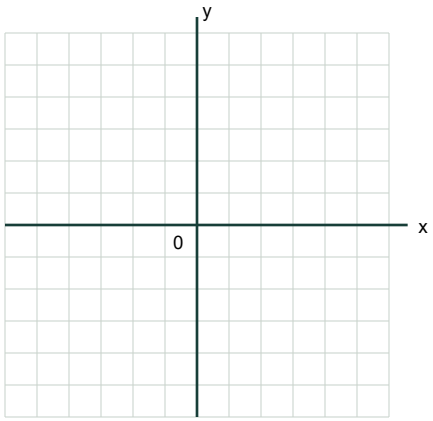
Plot (4, -5) and (4, -9). Find their distance.



Label the scale and units.

02 / Constructed response

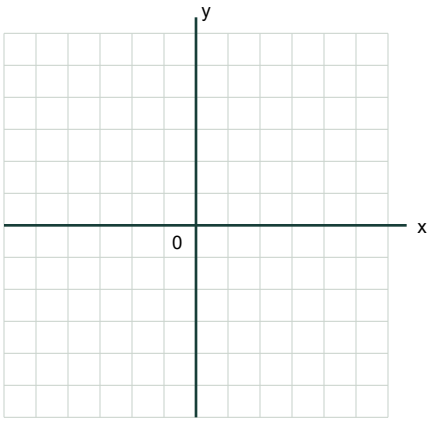
Plot (-8, 2) and (-10, 2). Find their distance.



Label the scale and units.

03 / Constructed response

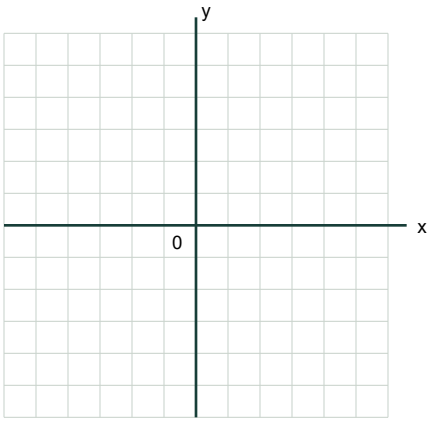
Plot (4, -2) and (4, -6). Find their distance.



Label the scale and units.

04 / Constructed response

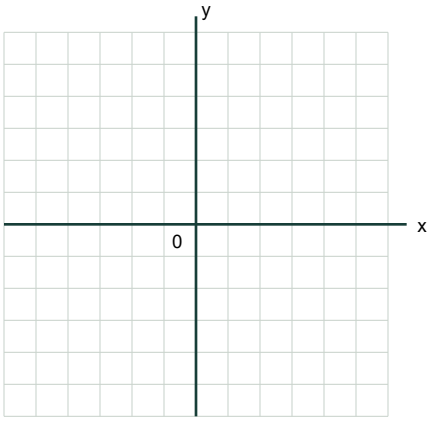
Plot (-5, 5) and (6, 5). Find their distance.



Label the scale and units.

05 / Constructed response

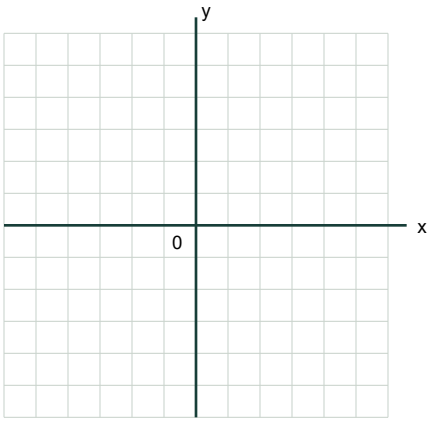
Plot (4, -7) and (4, 2). Find their distance.



Label the scale and units.

06 / Constructed response

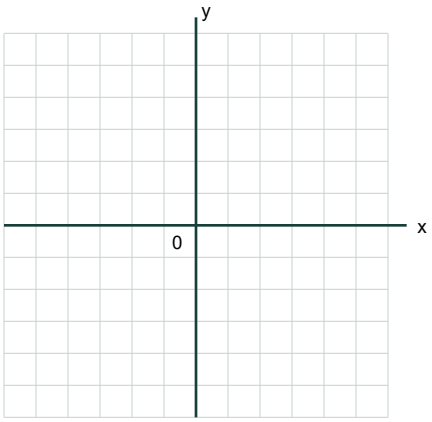
Plot (-7, 6) and (5, 6). Find their distance.



Label the scale and units.

07 / Constructed response

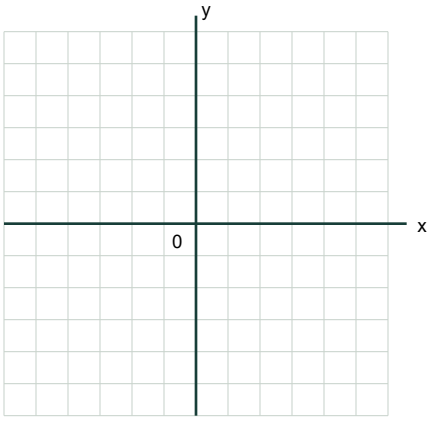
Plot (5, -3) and (5, 2). Find their distance. Explain why adding magnitudes works or does not work for these points.



Label the scale and units.

08 / Coordinate plotting and numeric entry

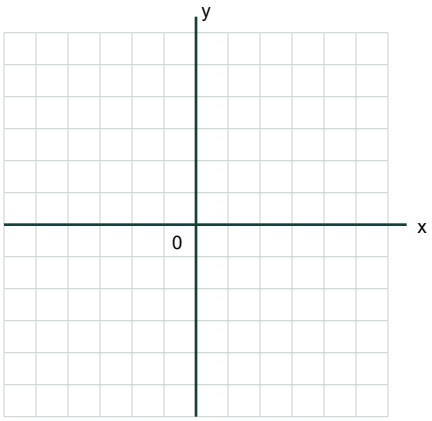
Plot A(-6, -5), B(3, -5), and C(-6, 5). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.



Label the scale and units.

09 / Coordinate plotting and numeric entry

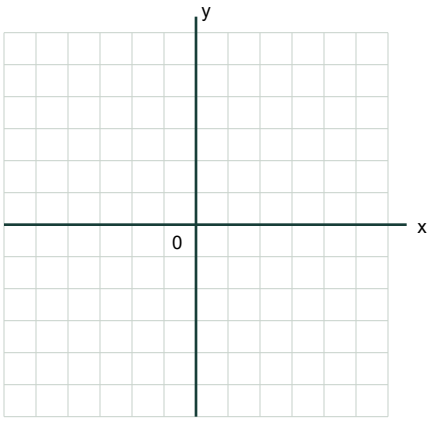
Plot A(-5, -6), B(3, -6), and C(-5, 6). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.



Label the scale and units.

10 / Constructed response

Plot (-5, 2) and (7, 2). Find their distance. Explain why adding magnitudes works or does not work for these points.



Label the scale and units.

Day 043 | Unit 03 | Answer key

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1. 4 units

Only one coordinate changes. Same side of 0: subtract magnitudes to get 4.

6.NS.C.8 | Constructed response

2. 2 units

Only one coordinate changes. Same side of 0: subtract magnitudes to get 2.

6.NS.C.8 | Constructed response

3. 4 units

Only one coordinate changes. Same side of 0: subtract magnitudes to get 4.

6.NS.C.8 | Constructed response

4. 11 units

Only one coordinate changes. Opposite sides of 0: 5 + 6 = 11.

6.NS.C.8 | Constructed response

5. 9 units

Only one coordinate changes. Opposite sides of 0: 7 + 2 = 9.

6.NS.C.8 | Constructed response

6. 12 units

Only one coordinate changes. Opposite sides of 0: 7 + 5 = 12.

6.NS.C.8 | Constructed response

7. 5 units

Only one coordinate changes. Opposite sides of 0: 3 + 2 = 5.

6.NS.C.8 | Constructed response

8. AB = 9 units; AC = 10 units

AB is horizontal: 6 + 3 = 9. AC is vertical: 5 + 5 = 10. Each segment crosses an axis.

6.NS.C.8 | Coordinate plotting and numeric entry

9. AB = 8 units; AC = 12 units

AB is horizontal: 5 + 3 = 8. AC is vertical: 6 + 6 = 12. Each segment crosses an axis.

6.NS.C.8 | Coordinate plotting and numeric entry

10. 12 units

Only one coordinate changes. Opposite sides of 0: 5 + 7 = 12.

6.NS.C.8 | Constructed response